

Oracle SQL Developer Web / Database Actions – Tools and Authentication Scenarios Overview

Oracle Database Actions, formerly known as SQL Developer Web, is a web-based interface running on Oracle REST Data Services and provides development, data tools, administration, and monitoring features for Oracle Database on-premises and Oracle Database Cloud services. Here all the authentication scenarios, prerequisites and the Database Actions services in the Cloud are presented.

Oracle Database Actions: Autonomous AI Database

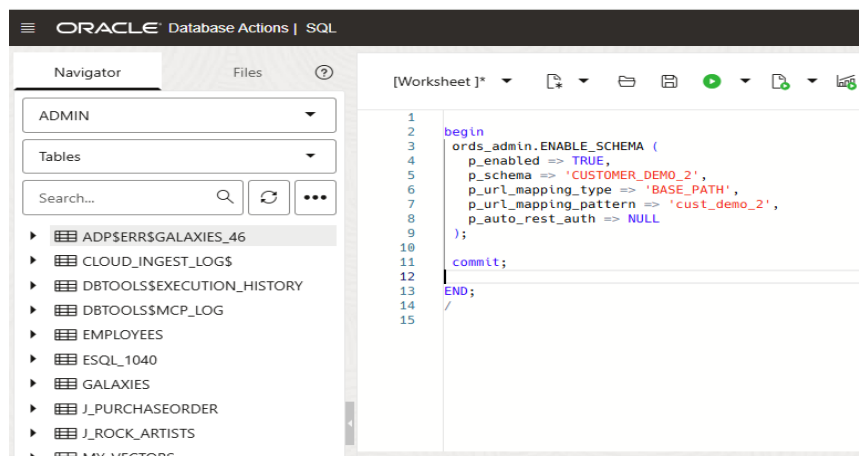
Prerequisites to access Database Actions:

In Oracle Autonomous Database, Database Actions is already built-in and the administrator user ADMIN is already enabled to authenticate. To enable a given schema user to authenticate, it has to be REST-enabled as follows, connected by ADMIN:

- From [User Administration](#) detail page:

```
-- REST ENABLE
BEGIN
  ORDS_ADMIN.ENABLE_SCHEMA(
    p_enabled => TRUE,
    p_schema => 'CUSTOMER_DEMO_2',
    p_url_mapping_type => 'BASE_PATH',
    p_url_mapping_pattern => 'customer_demo_2',
    p_auto_rest_auth => FALSE
  );
  -- ENABLE DATA SHARING
  C##ADP$SERVICE.DBMS_SHARE.ENABLE_SCHEMA(
    SCHEMA_NAME => 'CUSTOMER_DEMO_2',
    ENABLED => TRUE
  );
  commit;
END;
/
```

- From [SQL Worksheet](#) via PL/SQL code:



NB: enabling user via PL/SQL in SQL Worksheet allows you to choose a *schema_alias* name (*p_url_mapping_partner* API attribute) different from the schema name.

In order to authenticate to Database Actions, other than being REST-enabled, to have at least minimal operational capabilities the schema user also needs the followings:

- Non-zero value on Tablespace Quota (unless it is a read-only user);
- CONNECT, RESOURCE assigned roles

For example, connected as ADMIN user and checking the roles assigned to a schema user:

```
5
6 select * from dba_role_privs where grantee='WKSP_DATASOLPROV';
7
8
9
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULT SQL HISTORY TASK MONITOR

All rows fetched: 4 in 0.073 seconds

	GRANTEE	GRANTED_ROLE	ADMIN_OPTION	DELEGATE_OPTION	DEFAULT_ROLE	COMMON	INHERITED
1	WKSP_DATASOLPROV	CONNECT	NO	NO	YES	NO	NO
2	WKSP_DATASOLPROV	RESOURCE	NO	NO	YES	NO	NO

Roles like DWROLE and DB_DEVELOPER_ROLE (which grants the access to Data Studio tools and other developer features) are considered add-ons.

Sign-In to Oracle Database Actions

You can sign-in to Database Actions by connecting to a single default database or to multiple databases or by using a schema alias.

- Sign-in to DB Actions on ADB via ORDS Schema alias which is different from schema user name:

Sign-in

Path Advanced ^

datasolprov

Sign-in URL: https://ig5e60ebadfd95a-myatp23ai.adb.eu-frankfurt...

Username

wksp_datasolprov

Password

Sign in

or

Sign in with SSO

Here on Advanced → Path, enter the schema alias (“datasolprov”) and in the Username/password the database schema user and correspondent password. The ORDS alias configuration for this schema user (REST-enabled) is as follows (run as ADMIN and query DBA_ORDS_SCHEMA data dictionary view) :

```
select parsing_schema, pattern from dba_ords_schemas where parsing_schema like '%DATASOLPROV%';
```

5 OUTPUT DEBUG CONSOLE TERMINAL PORTS <u>QUERY RESULT</u> SQL HISTORY TASK MONITOR		
fetched: 1 in 0.044 seconds		
PARSING_SCHEMA	PATTERN	
WKSP_DATASOLPROV	datasolprov	

Looking at the PL/SQL script enabling WKSP_DATASOLPROV as REST-enabled schema we can notice the schema_alias as “datasolprov” (p_url_mapping_pattern PL/SQL API attribute):

```
ORDS_ADMIN.ENABLE_SCHEMA(  
  p_enabled => TRUE,  
  p_schema => 'WKSP_DATASOLPROV',  
  p_url_mapping_type => 'BASE_PATH',  
  p_url_mapping_pattern => 'datasolprov',  
  p_auto_rest_auth=> FALSE  
);
```

- Sign-in to DB Actions on ADB via ORDS Schema alias with the same name as schema user:

Sign-in

✓ You have been signed out

Path Advanced ^

Database mapping path and/or schema alias

Sign-in URL: <https://g5e60ebadf0d95a-myatp23ai.adb.eu-frankfurt...>

Username

mcp_usr

Password

Sign in

or

Sign in with SSO

Here there's no need to enter the schema alias in the Advanced → Path as the schema alias is equal to Database user Schema (not recommended as best practice). If we query the data dictionary view we can notice:



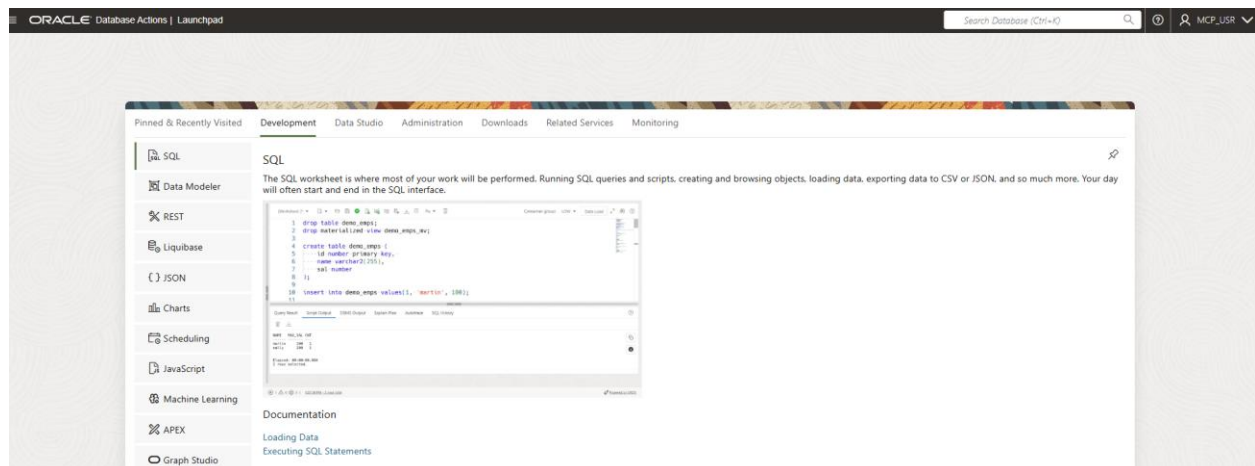
```
select parsing_schema, pattern from dba_ords_schemas where parsing_schema = 'MCP_USR';
```

S OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULT SQL HISTORY TASK MONITOR

fetches: 1 in 0.047 seconds

PARSING_SCHEMA	PATTERN
MCP_USR	mcp_usr

Once you connect to Database Actions (whatever the schema alias scenario) the main DB Actions page is as follows:



In which the main tools are the SQL Worksheet, Monitoring and REST. All of these options are allowed depending on the roles/privileges assigned, by ADMIN user, to current schema user.

Oracle Database Actions: Database Cloud Service/ExaDB/on-prem DB

*** Work in Progress ***